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Optimizing energy flow in advanced microgrids: a prediction-independent two-stage hybrid system approach

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Abstract
This paper presents a two-stage optimization framework for long-term energy management in microgrids, aiming to efficiently integrate various energy sources, storage systems, and consumption elements while addressing uncertainties in load demand and renewable generation. The framework consists of an offline optimization stage and an online optimization stage, each with distinct roles to balance long-term planning and real-time adaptability. In the offline stage, a robust two-stage mixed-integer linear programming (MILP) model is used to set annual targets for the state of charge (SoC) of energy storage systems. This stage applies a min-max-min approach to optimize for worst-case scenarios, establishing a cost-effective and reliable baseline plan that reduces dependency on conventional power sources and minimizes load deficits. The online stage, on the other hand, employs a new online convex optimization model that dynamically adjusts energy storage and dispatch decisions based on real-time data, allowing the microgrid to respond flexibly to fluctuations in demand and renewable generation. Simulation results using the Elia and North China datasets demonstrate the effectiveness of this two-stage approach. Offline optimization achieved up to 25% cost savings and reduced unmet demand by up to 99%, providing a stable foundation for efficient energy management. The online optimization stage further improved system responsiveness, minimizing reliance on backup generators and enhancing load reliability. This combined framework offers a comprehensive solution for optimizing microgrid performance, balancing predictive planning with real-time adaptability in complex, variable energy environments.

Keywords Hybrid storage system, Microgrid, Electric vehicle, Online convex optimization, Robust two-stage optimization

Introduction

Efficient energy flow management in modern microgrids has become a critical area of research due to the increasing complexity of integrating diverse energy sources and storage systems. Microgrids, which are localized energy systems capable of operating independently or in conjunction with the main grid, play a vital role in enhancing energy reliability, sustainability, and cost-efficiency. A typical advanced microgrid may consist of renewable energy sources, such as solar and wind, along with conventional power generation, hydrogen-battery storage systems, electric vehicles, and adaptable consumption patterns. This complexity necessitates sophisticated management and control strategies to ensure optimal performance while addressing economic and environmental concerns. The significance of optimizing energy flow in microgrids lies in its potential to contribute to a more resilient and sustainable energy infrastructure. The integration of hybrid energy storage systems, such as hydrogen and battery, alongside renewable energy sources, allows for a more flexible and efficient response to fluctuating energy demands and unpredictable supply conditions. Moreover, the rise of electric vehicles introduces additional variables that need to be managed effectively. An optimized microgrid system can not only reduce operational costs and improve energy efficiency but also minimize reliance on conventional energy sources, contributing to a reduction in carbon emissions. This highlights the pressing need for advanced optimization techniques that can accommodate the dynamic and unpredictable nature of modern energy systems. However, achieving optimal energy flow management in such a diverse setup presents several challenges. The primary challenges include the accurate modeling of storage systems, the handling of uncertainties in renewable energy output and load demand, and the development of effective decision-making algorithms that can operate in real time. Traditional optimization methods often rely heavily on predictive models, which may not adequately capture the inherent uncertainties of renewable energy generation and varying consumption patterns. Additionally, balancing operational costs while ensuring energy reliability in the face of these uncertainties is a demanding task. This paper addresses these challenges by proposing a comprehensive two-stage optimization framework. The suggested framework is prediction-independent, enabling more robust decision-making by setting annual state-of-charge targets for hydrogen storage systems offline and refining them in real time through advanced kernel regression techniques. A novel online convex optimization method is introduced to manage energy flows efficiently, while uncertainties are handled using a robust two-stage optimization strategy leveraging the column-and-constraint generation algorithm. The objective of this research is to provide a more resilient and cost-effective solution for energy management in modern microgrids, fulfilling the growing need for scalable and adaptable energy systems.

Motivation

The motivation for this paper arises from the pressing need to enhance the efficiency, reliability, and sustainability of modern energy systems. With the rapid growth of renewable energy integration, the traditional centralized energy management models are struggling to cope with the variability and uncertainty inherent in renewable sources. As microgrids gain prominence as localized solutions for flexible and resilient energy supply, there is a critical demand for advanced management strategies that can effectively handle the complexities of hybrid energy systems, including renewable sources, hydrogen-battery storage, and electric vehicles. Existing optimization approaches often rely on predictive models that may not adequately capture the dynamic nature of modern energy environments. This reliance on predictions can lead to inefficiencies and increased costs, especially when unexpected changes in load demand or renewable generation occur. Additionally, the economic pressures to reduce operational expenses and the environmental imperative to minimize carbon footprints further underline the necessity for innovative and prediction-independent optimization techniques. The motivation behind this research is to bridge the gap between theoretical optimization models and practical energy management needs. By introducing a two-stage, prediction-independent framework, this study aims to provide a more adaptable and robust solution that can better accommodate the uncertainties and complexities of advanced microgrids. The development of a semi-empirical hydrogen storage model, alongside a real-time refinement strategy, seeks to enhance the accuracy and efficiency of energy management. This research is driven by the desire to reduce operational costs, improve the reliability of energy supply, and support the transition to sustainable energy systems through the integration of novel optimization methodologies.

Literature review

Recent advancements in microgrid energy management have demonstrated the effectiveness of intelligent optimization techniques in enhancing system resilience, efficiency, and sustainability. A two-stage artificial intelligence (AI)-based framework was proposed in [1] to improve microgrid resilience by integrating a hybrid optimization algorithm with a backup energy management system. Similarly, adaptive control strategies based on hybrid intelligent algorithms were employed in [2] to boost smart grid efficiency.

In [3], a multifunctional energy management system was developed for microgrids that incorporates battery degradation and load variability to ensure system stability. A hierarchical multi-agent reinforcement learning model was presented in [4], leveraging trust-region policy optimization to coordinate operations across multi-energy, multi-microgrid systems. The integration of electric vehicles into hybrid microgrids was addressed in [5] through a stochastic framework that manages renewable energy variability and energy fluctuations.

To address demand and supply uncertainty, an enhanced Teaching-Learning-Based Optimization (TLBO) algorithm combined with support vector machine forecasting was utilized in [6]. Voltage and frequency control in interconnected microgrids were improved using a hierarchical deep learning model in [7], while a neuro-fuzzy approach was proposed in [8] to support real-time energy management decisions. A multi-objective optimization model targeting the integration of microgrids and energy hubs was developed in [9], and hybrid energy storage systems incorporating batteries and supercapacitors were analyzed comprehensively in [10].

The study in [11] introduced a techno-economic optimization model for hybrid energy systems with limited grid access, aimed at achieving net-zero emissions in New Zealand. Advanced data-driven approaches for photovoltaic (PV) forecasting were explored in [12], focusing on feature engineering and computational efficiency. An open-source tool, SAMA, for multi-objective optimization in solar-integrated microgrids was presented in [13].

In [14], a robust-stochastic optimization framework incorporating deep learning models of social behavior was proposed to evaluate microgrid resilience. The African Vultures Optimization Algorithm was applied in [15] to develop a resilient energy management system for residential microgrids. An artificial neural network (ANN) NARMA-L2 controller was employed in [16] to manage hybrid storage systems and enhance efficiency. A rolling dispatch strategy across multiple timescales was developed in [17] to coordinate dynamic and flexible energy sources.

Dual-layer energy management strategies for AC microgrids were investigated in [18], focusing on power control schemes. A comprehensive review of AI-based energy management systems in microgrids was conducted in [19], summarizing emerging trends and methodologies. A distributed model predictive control (MPC) framework was proposed in [20] to ensure robust coordination across multi-microgrid environments.

Local integration of renewable energy sources in microgrids was explored through a case study in Hurghada, Egypt, in [21]. To ensure data privacy in distributed systems, homomorphic encryption techniques were adopted in [22] for secure management of microgrid clusters. A hierarchical plug-and-play management framework for neighborhood microgrids was proposed in [23]. Coordinated operation of wind and solar generation with hybrid storage for grid stability was addressed in [24].

The latest developments in economic MPC for microgrids were reviewed in [25], offering insights into model design and real-time deployment. A stochastic optimization strategy was presented in [26] for autonomous PV microgrids under uncertain load conditions. Simulation models for advanced alkaline electrolyzers were developed in [27] to characterize their integration into energy systems, while a semi-empirical model validated for a 15 kW alkaline electrolyzer was proposed in [28] to assess performance under varying operating scenarios.

A prediction-free optimization framework for long-term microgrid energy management was developed in [29], integrating hydrogen and battery storage to improve system flexibility. Similarly, a long-term strategy for isolated microgrids utilizing hybrid seasonal-battery storage was proposed in [30], with a focus on system reliability and energy balancing.

Historical data sets were employed in [31] and [32] to enhance the realism of simulation-based microgrid studies, drawing from Belgian and Chinese grid operations, respectively. A two-stage distributed stochastic optimization method was introduced in [33] for green multi-energy ship microgrids, incorporating underwater noise constraints to enhance marine sustainability. In [34], a distributed hybrid-triggered observer-based secondary control method was developed to improve voltage regulation in multi-bus DC microgrids.

A spatiotemporal multi-task learning model was proposed in [35] to accurately predict electric vehicle charging demand, with an emphasis on explainability. Smart home energy management was optimized under uncertainty in [36] using a robust framework integrating PVs, batteries, EV charging, and demand response. The economic dispatch of solar-integrated systems was studied in [37] with attention to flexibility and reliability.

Battery investment and grid expansion were co-optimized in [38] to support integrated energy system planning under uncertain demand. A current-limiting scheme was proposed in [39] to improve protection coordination in electric vehicle-integrated distribution networks. Finally, a strategic investment model for large-scale energy storage was introduced in [40] to enhance the profitability of demand response aggregators in liberalized electricity markets.

Table 1 compares the reviewed studies based on optimization methods, energy sources, storage systems, and decision-making frameworks. Unlike previous works, our study introduces a novel two-stage optimization framework combining online convex

Table 1 Comparison of key aspects in microgrid energy management studies

Study	Focus/Methodology	Optimization techniques	Energy sources considered	Energy storage systems	Decision-Making framework
This paper	Prediction-independent and robust optimization for microgrid energy management	Two-stage optimization; kernel regression; online convex optimization	Renewable energy sources; conventional power generation	Hydrogen-battery storage; electric vehicles	Real-time adjustments; annual SoC targets
[1]	AI-enhanced system for microgrid resilience	Hybrid optimization algorithm	Not specified	Backup energy management system	AI-driven adaptability
[2]	Hybrid intelligent algorithms for energy management	Intelligent adaptive control strategies	Smart grid operations	Not specified	Adaptive control
[3]	Multifunctional energy management for microgrids	Not specified	Battery degradation; load adjustments	Battery storage	Stability optimization
[4]	Hierarchical multi-agent reinforcement learning for optimization	Trust-region optimization techniques	Multi-energy, multi-microgrid systems	Not specified	Collaborative optimization
[5]	Stochastic approach for integrating EVs in hybrid microgrids	Stochastic optimization	Renewable energy; electric vehicles	Not specified	Framework for future-ready solutions
[6]	TLBO algorithm for optimal energy management	Teaching-Learning-Based Optimization; support vector machine forecasting	Not specified	Hybrid microgrids	Forecasting-based optimization
[7]	Hierarchical deep learning for voltage and frequency control	Deep learning frameworks	Interconnected microgrid systems	Not specified	Stability and efficiency optimization
[8]	Neural-fuzzy optimization for energy management	Neural-fuzzy techniques	Not specified	Not specified	Real-time decision-making
[9]	Multi-objective optimization for energy hubs	Algorithmic multi-objective optimization	Microgrid and power system integration	Not specified	Balancing multiple objectives
[10]	Hybrid energy storage systems analysis	Not specified	Battery banks; supercapacitors	Hybrid energy storage systems	Energy optimization
[11]	Techno-economic optimization for hybrid energy systems	Techno-economic optimization	Not specified	Hydrogen storage	Net-zero carbon emissions
[12]	PV forecasting using advanced feature engineering	Feature engineering; fast computing	PV microgrids	Not specified	Data-driven modeling
[13]	SAMA tool for microgrid optimization	Multiple objective optimization	Solar energy solutions	Not specified	Optimization tool
[14]	Multi-objective robust-stochastic optimization	Robust-stochastic optimization	Multi-energy microgrids	Battery storage	Deep learning for social behavior simulation
[15]	African Vultures Optimization Algorithm for residential microgrids	African Vultures Optimization Algorithm	Autonomous residential microgrids	Not specified	Resilient operation
[16]	ANN NARMA-L2 controller for hybrid energy storage	ANN NARMA-L2 control techniques	Not specified	Hybrid energy storage systems	Control-based optimization
[17]	Multi-timescale rolling optimization dispatch	Rolling optimization	Integrated energy systems	Hybrid storage	Coordinating multiple energy sources

Table 1 (continued)

Study	Focus/Methodology	Optimization techniques	Energy sources considered	Energy storage systems	Decision-Making framework
[18]	Dual-layer energy management in AC microgrids	Power control strategies	AC microgrids	Not specified	Reliability enhancement
[19]	Review of computational intelligence approaches	Computational intelligence review	Microgrid energy management	Not specified	AI-driven strategies overview
[20]	Robust energy management for multi-microgrid systems	Distributed dynamic tube model predictive control	Multi-microgrid systems	Not specified	Reliable and efficient energy distribution
[21]	Case study of renewable energy integration in Hurghada, Egypt	Case study analysis	Renewable energy sources	Not specified	Local renewable energy adoption
[22]	Privacy-preserving energy management for networked microgrids	Homomorphic encryption	Networked microgrid clusters	Not specified	Data security in optimization
[23]	Hierarchical energy management with plug-and-play capability	Plug-and-play capability	Neighborhood microgrids	Not specified	Adaptable solutions
[24]	Coordinated operation of wind and solar in grid-connected microgrids	Hybrid energy storage management	Wind; solar energy	Grid-connected microgrids	System stability
[25]	Economic model predictive control for microgrid optimization	Model predictive control	Microgrid optimization	Not specified	Predictive control techniques
[26]	Stochastic energy management for autonomous PV microgrids	Stochastic optimization	Autonomous PV microgrids	battery storage	Handling uncertainties

optimization with kernel regression in a prediction-independent manner. It supports a wide range of energy sources, including hydrogen-battery systems and electric vehicles. Our approach features offline annual state-of-charge (SoC) targets for hydrogen storage, with real-time adjustments via kernel regression. This enhances flexibility and accuracy in energy flow management. Simulation results show significant reductions in operational costs (~75%) and load deficits (~96%) compared to conventional methods. The proposed robust strategy effectively manages uncertainties using a column-and-constraint generation (C&CG) algorithm, offering a clear advancement over existing solutions.

Research gap

Despite significant advancements in microgrid energy management, there are notable research gaps that need to be addressed for enhancing the efficiency and reliability of these systems. Many existing studies focus on specific aspects of optimization, such as integrating AI, hybrid intelligent algorithms, or stochastic approaches. However, they often lack a comprehensive framework that combines prediction-independent methods, robust optimization, and effective handling of uncertainties in load demand and renewable energy output. Additionally, while various studies explore different energy sources and storage systems, there is a scarcity of research that integrates a diverse range of sources, including hydrogen-battery storage and electric vehicles, within a single optimization framework. Furthermore, conventional methods often fail to provide flexible and accurate real-time decision-making, especially in complex and dynamic microgrid environments. This paper addresses these research gaps by introducing a novel two-stage optimization framework that combines prediction-independent methods with robust optimization techniques. By incorporating advanced kernel regression for real-time adjustments and a semi-empirical approach for hydrogen storage system modeling, the study provides a more flexible and accurate decision-making process. The robust two-stage optimization strategy effectively manages uncertainties using the column-and-constraint generation (C&CG) algorithm, leading to substantial improvements in operational costs and load deficits. This comprehensive approach integrates diverse energy sources and storage systems, demonstrating significant advancements over existing methodologies and highlighting the necessity of such integrated frameworks in modern microgrid systems.

Contribution

This study introduces the following key contributions to the field of microgrid energy management:

1. **Development of a Hybrid Energy Storage Management Framework**: We present a comprehensive energy storage system that integrates hydrogen and battery technologies. A semi-empirical model for hydrogen storage is proposed to improve efficiency estimation under varying operational conditions.
2. **Prediction-Independent Optimization with Real-Time Adaptation**: A novel two-stage optimization framework is introduced, combining robust offline planning (via the column-and-constraint generation algorithm) with online convex optimization and kernel regression. This approach eliminates the need for forecasting and allows dynamic SoC reference tracking for hydrogen storage, enabling flexible and accurate decision-making under uncertainty.
3. **Demonstration of Practical Effectiveness via Comparative Simulations**: The proposed approach is validated using datasets from Elia (Belgium) and North China. Results show up to 25% operational cost savings and 99% load deficit reduction compared to conventional methods, demonstrating the framework's ability to enhance system reliability and scalability across diverse environments.

Overall, this research contributes to the ongoing development of resilient, cost-effective, and environmentally friendly energy management solutions, paving the way for more efficient integration of renewable energy and advanced storage systems in modern microgrids. Figure 1 illustrates a proposed microgrid energy management schematic, showcasing various components and their interactions. The central component is the cloud-based control system, which acts as the brain of the microgrid management system, coordinating and optimizing energy flow among all connected components. Wind turbines, connected via a purple arrow to the cloud-based control system, feed wind energy into the microgrid, contributing to renewable energy generation. Industrial loads, represented by a red arrow, indicate industrial energy consumption, one of the variable loads in the microgrid. Solar panels, connected via a yellow arrow, provide solar energy, another renewable source integrated into the microgrid. Electric vehicles, linked via a blue arrow, act as both loads (when charging) and storage units (vehicle-to-grid services). Hydrogen storage, connected via a green arrow, offers long-term energy storage solutions, helping balance supply and demand. The grid connection, also connected via a green arrow, allows for energy exchange with the main grid, ensuring reliability and stability. Battery storage, represented by another green arrow, provides short-term

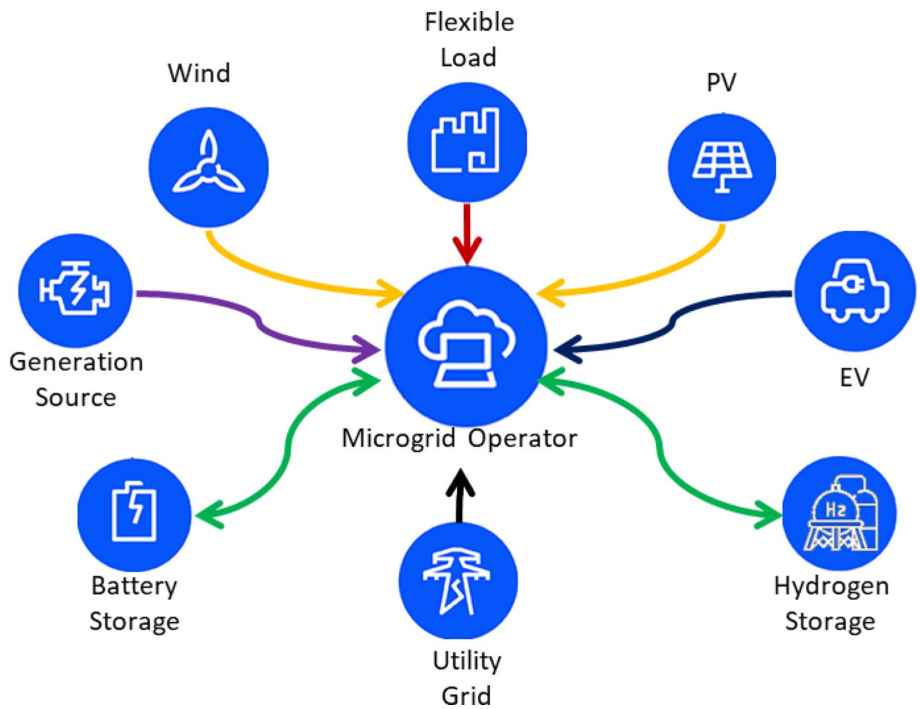

Fig. 1 Schematic of proposed microgrid energy management

energy storage, aiding in demand response and grid stability. Combined Heat and Power (CHP) units, linked via a purple arrow, offer efficient energy generation by producing both electricity and heat. Energy sources such as wind, solar, and CHP units feed energy into the microgrid, managed by the central control system to optimize generation and minimize costs. Energy storage solutions, including batteries and hydrogen storage, store excess energy generated by renewables or during low demand periods and provide energy during high demand or low generation periods. Energy consumption from industrial loads and electric vehicles is distributed based on real-time demand and supply conditions, with the central control system ensuring efficient and balanced energy distribution. The microgrid can import energy from the grid during shortages or export excess energy, enhancing the overall stability and reliability of the energy system. The central control system utilizes advanced optimization techniques, such as kernel regression and convex optimization, for real-time decision-making, ensuring efficient energy management, cost reduction, and load balancing. Robust optimization strategies handle uncertainties in renewable output and load demand, enhancing the resilience and reliability of the microgrid. This schematic represents a sophisticated energy management system for a microgrid, integrating various energy sources, storage options, and loads, with the cloud-based control system playing a crucial role in optimizing energy flow, managing uncertainties, and ensuring efficient and reliable operation.

Paper organization

The proposed modeling section discusses the integration of hydrogen storage, batteries, electric vehicles, diesel generators, and renewable energy sources, along with the established objective function. The two-stage optimization section elaborates on the offline and online optimization methodologies employed to enhance operational efficiency. The simulation results section presents the outcomes of the optimization processes, highlighting key performance metrics for both datasets. The conclusions and suggestions section summarizes the findings and provides recommendations for future research and practical implementations in microgrid management.

Proposed model

In this section, proposed modeling for energy management including modeling of renewable resources, loads, hybrid-energy storage systems, electric vehicles, equality and inequality constraints are presented.

Hydrogen energy storage model

A hydrogen storage system typically includes essential units like electrolyzers, hydrogen tanks, and fuel cells. Electrolyzers convert electrical power to chemical energy by producing hydrogen and oxygen; here, we examine the commonly used alkaline water type. Stored hydrogen can be converted back into electricity by fuel cells, with a focus on proton exchange membrane fuel cells (PEMFCs). Additional equipment, such as compressors, cooling, and control systems, is not part of the modeling scope in this work. The electrochemical performance of an electrolyzer is represented by its polarization curve, which outlines how current and voltage interact. To include the effects of both temperature and pressure on the electrolyzer's thermodynamic and reaction processes, we utilize a model that combines the well-recognized frameworks developed by studies [27] and [28].

$$U_{\mathrm{cell}}^{\mathrm{E}} = U_{\mathrm{rev}} + [(r_1 + d_1) + r_2 \cdot \theta + d_2 \cdot P] \cdot \frac{i}{A} + s \cdot log\left[\left(t_1 + \frac{t_2}{\theta} + \frac{t_3}{\theta^2}\right) \cdot \frac{i}{A} + 1\right] \tag{1}$$

In the electrolyzer, U_{rev} refers to the reversible voltage, and $U_{\mathrm{cell}}^{\mathrm{E}}$ represents the cell voltage. The variables θ and P denote temperature and pressure, respectively. The electrode's current is labeled i, and its effective area is A. The constants, including $r_1, r_2, d_1, d_2, t_1, t_2, t_3$, and s, are determined based on experimental observations. Faraday efficiency quantifies the actual hydrogen output compared to its theoretical yield. r_1 and r_2 represent the slope and intercept of the segmented linear model for hydrogen production efficiency. These parameters vary by electrolyzer type and are derived from experimental fitting, often using data from [27] and [28]. They can be generalized for similar industrial alkaline electrolyzers. d_1 and d_2 correspond to coefficients in the polarization curve fitting equation of the PEMFC, capturing activation and ohmic losses. These values depend on membrane material and temperature, but ranges are typically consistent across PEMFC systems of a given class. t_1, t_2, t_3 are coefficients used in the Faraday efficiency approximation (Eq. 2). They describe nonlinear efficiency loss due to current density and are derived from experimental data. While they are system-specific, similar patterns appear across commercial systems. The s represents the slope in a piecewise-linear charging/discharging cost function. For alkaline electrolyzers, this efficiency commonly lies within an 85–95% range, with temperature significantly impacting these values. In this study, we utilize a four-parameter model to represent Faraday efficiency, as indicated in Eq. (2).

$$\eta_F = \left(\frac{(i/A)^2}{f_1 + f_2 \cdot \theta + (i/A)^2} \right) \cdot (f_3 + f_4 \cdot \theta) \tag{2}$$

Faraday efficiency, denoted as η_{F}, represents a measure of conversion efficiency. The coefficients $f_1 - f_4$ represent the nonlinear impact of current density on Faraday efficiency and are fitted based on experimental data. In this study, values are adopted from Sánchez et al. [28], which are suitable for industrial alkaline electrolyzers under typical operating conditions. While the polynomial structure is widely applicable, coefficient values should be re-calibrated if applied to different electrolyzer designs or environmental conditions. Faraday's law outlines the definition for the rate of hydrogen generation, as expressed in Eq. (3). Meanwhile, the efficiency of the charging process is represented by Eq. (4).

$$h^{\mathrm{c}} = 3600 \cdot \frac{\eta_{\mathrm{F}} \cdot M \cdot i \cdot N}{2F} \tag{3}$$

$$\eta^{\mathrm{H,c}} = \frac{h^{\mathrm{c}} \cdot \mathrm{LHV}}{P_{\mathrm{Stack}}} = 3600 \cdot \frac{\eta_{\mathrm{F}} \cdot M \cdot \mathrm{LHV}}{2F \cdot U_{\mathrm{cell}}} \tag{4}$$

In this context, h^{c} represents the rate at which hydrogen is produced by the electrolyzer. The molar mass of hydrogen is denoted as M. The constant F, known as Faraday's constant, is equal to 96,485 coulombs per mole. The number of cells within the stack is indicated by N. Additionally, the lower heating value (LHV) of hydrogen is specified as 33.33 kilowatt-hours per kilogram. To enable manageable optimization of dispatch, we employ a segmented linear model for the generation of hydrogen [29]. The behavior of a Proton Exchange Membrane Fuel Cell (PEMFC) is usually represented through the equivalent circuit model introduced by [29]. The voltage across the cell, denoted as $U_{\mathrm{cell}}^{\mathrm{F}}$, is expressed by Eqs. (5)–(9). The constants a_1, a_2, and B are derived from experimental studies of low-temperature PEMFCs using Nafion membranes [30]. These values are representative of commonly used commercial systems but may require adjustment for other membrane materials, operating pressures, or temperatures. Users should refer to manufacturer datasheets or conduct calibration for different configurations. This voltage is essentially the Nernst open circuit voltage E_{Nernst} reduced by three distinct irreversible losses: activation losses U_{act}, concentration losses U_{con}, and ohmic losses U_{ohmic}.

$$U_{\mathrm{cell}}^{\mathrm{F}} = E_{\mathrm{Nernst}} - U_{\mathrm{act}} - U_{\mathrm{ohmic}} - U_{\mathrm{con}} \tag{5}$$

$$E_{\mathrm{Nernst}} = \frac{1}{2F} \left[\Delta G - \Delta S (\theta - \theta_{\mathrm{ref}}) + R \cdot \theta \left(log (P_{H_2}) + \frac{log (P_{O_2})}{2} \right) \right] \tag{6}$$

$$U_{\mathrm{act}} = a_1 + a_2 \cdot \theta + a_3 \cdot \theta \cdot \log (C_{\mathrm{O_2}}) + a_4 \cdot \theta \cdot \log (i) \tag{7}$$

$$U_{\mathrm{ohmic}} = i \cdot R_{\mathrm{ohmic}} = i (r_M \cdot l/A + R_c) \tag{8}$$

$$U_{\mathrm{con}} = B \cdot \log \left(1 - \frac{J}{J_{\max}} \right), J = \frac{i}{A} \tag{9}$$

Table 2 summarizes key electrochemical parameters used in modeling the alkaline electrolyzer and PEM fuel cell components of the hybrid storage system. The constants are based on widely accepted empirical models and literature sources. While the equations are general, parameter values are calibrated from experimental data and may require adjustment when applied to systems with different materials, operating conditions, or design specifications.

ΔG stands for the Gibbs free energy, and ΔS reflects the variation in entropy. The gas constant, represented as R, is 8.314 J/(K·mol). The terms $P_{\mathrm{H_2}}$ and $P_{\mathrm{O_2}}$ correspond to the partial pressures of hydrogen and oxygen, respectively. The reference temperature, noted as T_{ref}, is 298.15 K. $C_{\mathrm{O_2}}$ represents the oxygen concentration at the surface of the cathode catalyst. The electrolyte membrane's resistivity is indicated by r_M, while l represents its thickness. B is the coefficient for concentration overpotential. J indicates the current density, with $J_{\max}$ denoting its highest value. The constants a_1, a_2, a_3 and a_4 can be obtained from experimental observations. The rate at which hydrogen is consumed is determined by Faraday's law, as expressed in Eq. (10). Additionally, the efficiency of the discharge process can be represented by Eq. (11).

$$h^{\mathrm{d}} = 3600 \cdot \frac{M \cdot i \cdot N}{2F} \tag{10}$$

$$\eta^{\mathrm{H,d}} = \frac{P_{\mathrm{Stack}}}{h^{\mathrm{d}} \cdot \mathrm{HHV}} = \frac{2F \cdot U_{\mathrm{cell}}}{3600M \cdot \mathrm{HHV}} \tag{11}$$

The variable h^{d} represents the rate at which hydrogen is utilized by the Proton Exchange Membrane Fuel Cell (PEMFC). The higher heating value (HHV) of hydrogen is quantified as 39.4 kWh per kilogram. We utilize a segmented linear model to estimate hydrogen usage. The model for hydrogen storage is detailed in equations (12) to (18). Equation (12) establishes the connection among the state of charge (SoC), charging power, and discharging power. Equation (13) sets boundaries for the SoC of the hydrogen storage system. Equation (14) ensures that the hydrogen storage maintains a stable energy state across multiple cycles. Equation (15) through (17) articulate the structured approach for representing charging and discharging processes using piecewise linear functions. Finally, Eq. (18) imposes restrictions on the power levels for charging and discharging hydrogen storage.

$$\text{Constraints} : \forall\, t \in \boldsymbol{\Omega}_T \forall\, p \in \boldsymbol{\Omega}_P \tag{12}$$

$$E^{\mathrm{H}}_{t+1} = E^{\mathrm{H}}_t + \Delta\, t \left(h^{\mathrm{c}}_t - h^{\mathrm{d}}_t\right) - E^{\mathrm{H,L}}_t$$

$$\underline{E}^{\mathrm{H}} \leq E^{\mathrm{H}}_t \leq \bar{E}^{\mathrm{H}} \tag{13}$$

Table 2 Description and sources of electrochemical model parameters

Parameter	Description	Unit	Value used	Notes
$\boldsymbol{f}_1 - \boldsymbol{f}_4$	Faraday efficiency coefficients	–	From [28]	Empirically fitted for alkaline electrolyzers
$\boldsymbol{a}_1, \boldsymbol{a}_2$	Activation loss constants	V, A^{-1}	From [30]	Valid for Nafion-based PEMFC at standard conditions
$\boldsymbol{B}$	Concentration loss decay factor	V	From [30]	Pressure- and design-dependent; may require tuning

$$E_T^{\mathrm{H}} \geq E_0^{\mathrm{H}} \tag{14}$$

$$h_t^{\mathrm{c}} = \sum_p \left(A_p P_{p,t}^{\mathrm{H,c}} + B_p z_{p,t}^{\mathrm{c}}\right), h_t^{\mathrm{d}} = \sum_p \left(C_p P_{p,t}^{\mathrm{H,d}} + D_p z_{p,t}^{\mathrm{d}}\right) \tag{15}$$

$$P_t^{\mathrm{H,c}} = \sum_p P_{p,t}^{\mathrm{H,c}}, P_t^{\mathrm{H,d}} = \sum_p P_{p,t}^{\mathrm{H,d}} \tag{16}$$

$$\sum_p z_{p,t}^{\mathrm{c}} = 1, \sum_p z_{p,t}^{\mathrm{d}} = 1 \tag{17}$$

$$\underline{p}_p^{\mathrm{H}} z_{p,t}^{\mathrm{c}} \leq P_{p,t}^{\mathrm{H,c}} \leq \bar{P}_p^{\mathrm{H}} z_{p,t}^{\mathrm{c}}, 0 \leq P_{p,t}^{\mathrm{H,d}} \leq \bar{P}_p^{\mathrm{H}} z_{p,t}^{\mathrm{d}} \tag{18}$$

In this context, $\mathbf{\Omega}_T$ and $\mathbf{\Omega}_S$ denote the collections of time intervals and parameter divisions, respectively. The variables $P_t^{\mathrm{H,c}}$, $P_t^{\mathrm{H,d}}$, and E_t^{H} represent the charge power, discharge power, and state of charge (SoC) associated with hydrogen storage systems. The SoC can be quantified either by assessing the mass of hydrogen or by expressing it as a fraction of the total rated capacity. The variable $E_t^{\mathrm{H,L}}$ represents the hydrogen demand from processes like fertilizer production and steel manufacturing. The parameters $\underline{E}^{\mathrm{H}}$ and $\bar{E}^{\mathrm{H}}$ define the minimum and maximum limits of the state of charge (SoC), while $\underline{p}^{\mathrm{H}}$ and $\bar{P}^{\mathrm{H}}$ establish the lower and upper thresholds for power output. The minimum charging power is determined by the electrolyzer's operational baseline, which usually falls between 15% and 20% of its rated capacity. A_p and B_p represent the gradient and the constant term of the segmented linear charging profiles. Similarly, C_p and D_p denote the gradient and constant term for the segmented linear discharging profiles. The variables $z_{p,t}^{\mathrm{c}}$ and $z_{p,t}^{\mathrm{d}}$ are binary indicators associated with the piecewise linear functions.

Battery energy storage model

The constraints presented in Eqs. (19) and (23) outline the limitations associated with battery usage, which closely resemble those imposed on hydrogen storage systems. Both storage technologies share common constraints, including those governing charging and discharging rates as well as the state of charge (SoC). Nonetheless, it is crucial to highlight that battery efficiency is assumed to remain constant, there is no established minimum charging power threshold specified in Eq. (22), and the self-discharge rate must be taken into account in Eq. (19). To ensure that the battery cannot be charged and discharged at the same time, a binary variable δ_t^b is introduced. To capture the long-term effects of repeated charging and discharging, a battery degradation cost is included in the model. Battery aging is modeled as a function of discharged energy, which serves as a proxy for cycling activity (23). where c_{deg} is the aging cost coefficient (e.g., USD/MWh), derived based on the battery's unit cost, expected life in equivalent full cycles, and capacity. This linear approximation is computationally efficient and widely used in long-term planning models.

$$E_{t+1}^{\mathrm{B}} = (1 - \varepsilon\,\Delta\,t)E_t^{\mathrm{B}} + \Delta\,t\left(\eta^{\,\mathrm{B,c}} P_t^{\mathrm{B,c}} - P_t^{\mathrm{B,d}}/\eta^{\,\mathrm{B,d}}\right) \tag{19}$$

$$\underline{E}^{\mathrm{B}} \leq E_t^{\mathrm{B}} \leq \bar{E}^{\mathrm{B}} \tag{20}$$

$$E_T^{\mathrm{B}} \geq E_0^{\mathrm{B}} \tag{21}$$

$$0 \leq P_t^{\mathrm{B,c}} \leq \delta_t^b \bar{P}^{\mathrm{B}},\ 0 \leq P_t^{B,d} \leq (1-\delta_t^b)\bar{P}^B \tag{22}$$

$$C_t^{\mathrm{aging}} = c_{\mathrm{deg}} \cdot \sum_t P_t^{\mathrm{dis}} \cdot \Delta t \tag{23}$$

Electric vehicle model

The equations from (24) to (35) illustrate how electric vehicles are integrated into the energy management framework of a microgrid. The net energy associated with charging and discharging the battery in electric vehicles, denoted as $e_t^{net\_ev}$, is presented for time *t*, taking into account the efficiency of each battery as described in relation (24). In this context, $p_t^{dis\_ev}$ and $p_t^{ch\_ev}$ represent the power associated with discharging and charging the electric vehicle batteries during time *t*, respectively. Additionally, η^{ev} reflects the charging efficiency of these electric vehicle batteries. Equation (25) illustrates the power flow to and from the network by electric vehicles during time *t*. This power is determined by the battery's charging and discharging rates. The charging and discharging conditions of these vehicles are detailed in Eq. (26). Additionally, $I_t^{dis\_ev}$ and $I_t^{ch\_ev}$ represent the discharge and charge conditions of electric vehicles during time *t*, respectively. The variable n_t^{ev} denotes whether the electric vehicles are connected to the network; a value of 1 indicates a connection, while a value of 0 indicates disconnection. Equations (27) and (28) delineate the constraints related to the charging and discharging capabilities of electric vehicles. The symbols $\underline{p}^{ch\_ev}$ and $\overline{p}^{ch\_ev}$ denote the minimum and maximum charging power levels of these vehicles, while $\underline{p}^{dis\_ev}$ and $\overline{p}^{dis\_ev}$ indicate their minimum and maximum discharging power levels. Additionally, Eq. (29) illustrates the energy state of the electric vehicle battery e_t^{ev}, with dr_t^{ev} signifying the energy requirements of electric vehicles at the specified time *t*. Equation (30) illustrates the constraints on the energy capacity of batteries in electric vehicles, which must remain within specified minimum $\underline{E}_t^{ev}$ and maximum $\overline{e}_t^{ev}$ limits. In Eqs. (31) and (32), the net energy involved in charging and discharging the electric vehicle battery is analyzed, along with the energy available in the battery at the start and end of the specified time period. Equation (33) through (35) illustrate the cost curves associated with charging and discharging electric vehicle batteries. In this context, c_t^{ev} represents the operational expenses of electric vehicles, while β_b^{ev} denotes the slope of the *k*th segment used for linear approximation of the cost curve. Additionally, $p_{k,t}^{ev}$ refers to the power associated with the *k*th segment for electric vehicles. Lastly, $\overline{p}_{k,t}^{ev}$ indicates the maximum power allocated for the *k*th segment during linearization, where *k* serves as the index for the segments of the curve, and *K* represents the overall set of these segments. In (33), $\beta_{k,t}^{\mathrm{ch}}$ represents the price of purchasing electricity (for charging), while $\beta_{k,t}^{\mathrm{dis}}$ represents the revenue rate for selling electricity back to the grid (via discharging). The net EV cost c_t^{ev} therefore reflects the actual financial impact of EV participation in the energy market, accounting for both inflow (cost) and outflow (benefit). In (34), the segmented power variables are separated into charging ($p_{k,t}^{\mathrm{ev,ch}}$) and discharging ($p_{k,t}^{\mathrm{ev,dis}}$) components, both defined as non-negative values. This ensures consistency with the net energy change equation (Eq. 24), where discharging contributes positively and charging reduces battery energy. The summation over all segments allows piecewise linear approximation of costs or constraints related to EV energy flow.

$$e_t^{net\_ev} = \eta_{dis}^{ev} p_t^{dis\_ev} - \frac{1}{\eta_{ch}^{ev}} p_t^{ch\_ev} \ \forall t \in T \tag{24}$$

$$p_t^{ev} = I_t^{dis\_ev} p_t^{dis\_ev} - I_t^{ch\_ev} p_t^{ch\_ev} \ \forall t \in T \tag{25}$$

$$I_t^{dis\_ev} + I_t^{ch\_ev} = n_t^{ev} \ \forall t \in T, n_t^{ev} \in \{0,1\} \tag{26}$$

$$I_t^{ch\_ev} \underline{p}^{ch\_ev} \le p_t^{ch_{ev}} \le I_t^{ch_{ev}} \overline{p}^{ch\_ev} \ \forall t \in T \tag{27}$$

$$I_t^{dis\_ev} \underline{p}^{dis\_ev} \le p_t^{dis_{ev}} \le I_t^{dis_{ev}} \overline{p}^{dis\_ev} \ \forall t \in T \tag{28}$$

$$e_t^{ev} = e_{t-1}^{ev} - e_t^{net_{ev}} - (1 - n_t^{ev})\, dr_t^{ev} \ \forall t \in T \tag{29}$$

$$\underline{E}_t^{ev} \le e_t^{ev} \le \overline{e}_t^{ev} \ \forall t \in T \tag{30}$$

$$e_{t=1}^{ev} = e_{t=24}^{ev} \tag{31}$$

$$e_{t=1}^{net\_ev} = e_{t=24}^{net\_ev} \tag{32}$$

$$c_t^{\mathrm{ev}} = n_t^{\mathrm{ev}} \sum_{k=1}^{K} \left(\beta_{k,t}^{\mathrm{ch}} \cdot p_{k,t}^{\mathrm{ev,ch}} - \beta_{k,t}^{\mathrm{dis}} \cdot p_{k,t}^{\mathrm{ev,dis}} \right) \tag{33}$$

$$n_t^{ev} \left(e_t^{ev} - e_{t-1}^{ev} \right) \ge -\sum_{k=1}^{K} p_{k,t}^{ev,ch}, \quad n_t^{ev} \left(e_t^{ev} - e_{t-1}^{ev} \right) \le \sum_{k=1}^{K} p_{k,t}^{ev,dis} \ \forall t \in T \tag{34}$$

$$0 \le p_{k,t}^{ev} \le \overline{p}_{k,t}^{ev} \ \forall t \in T \tag{35}$$

Renewable energy model

The uncertain variable g_t represents the renewable generation at time t, which may deviate from its nominal (forecasted) value $\hat{g}_t$. To represent these variations in a mathematically tractable form, an uncertainty set $\mathcal{U}$ is defined as follows:

$$\mathcal{U} := \left\{ g \in \mathbb{R}^{|T|} : g_t = \hat{g}_t + \xi_t \cdot \delta_t, \ |\delta_t| \le \bar{\delta}_t, \ \sum_{t \in T} \frac{|\delta_t|}{\bar{\delta}_t} \le \Gamma \right\} \tag{36}$$

Here, $\hat{g}_t$ is the forecasted renewable output at time t, and $\bar{\delta}_t$ denotes the maximum allowable deviation from the forecast. The parameter Γ is a user-defined uncertainty budget that limits the total deviation over the entire horizon. This structure allows the model to consider worst-case scenarios while avoiding overly conservative solutions. The uncertainty set $\mathcal{U}$ is applied in the robust constraint formulation in Eq. (36) and integrated into the iterative column-and-constraint generation (C&CG) process, where new uncertainty realizations are identified and added to the master problem until convergence.

Problem formulation

The goal outlined in Eqs. (37)–(38) focuses on reducing the overall expenses of the system. These expenses include the generation costs associated with the diesel generator, penalties incurred from load reductions during island mode, and the operational expenses of the hybrid battery energy storage (H-BES). Equations (39) and (40) establish the limits for power output and the permissible ramp rates of the diesel generator. In this context, c^{L} and P_t^{L} represent the costs associated with load reduction and the corresponding power levels for load curtailment. c^{D} and P_t^{D} denote the pricing of fuel and the output of the diesel generator. Meanwhile, c^{B} and c^{H} indicate the marginal costs related to battery and hydrogen storage discharges. The variable Δt refers to the specified time duration. The limitations set by Eqs. (41) and (42) govern both the reduction of load and the use of renewable energy sources that can be controlled. Meanwhile, Eq. (43) places restrictions on the electricity imported from the primary grid. The equation that represents the balance of power is provided in Eq. (44). Furthermore, the conditions pertaining to the charging and discharging of battery and hydrogen storage systems have been simplified and excluded from the model due to the fulfillment of necessary criteria, specifically that the cost of discharging (represented as '+') surpasses that of charging (represented as '0'). Additionally, the constraints regarding power flow have been disregarded in the dispatch model, as microgrid systems are typically engineered to ensure high reliability and significant redundancy. The lower and upper power limits for the diesel generator are denoted as $\underline{p}^{\mathrm{D}}$ and $\bar{P}^{\mathrm{D}}$, respectively. The parameters RD^{D} and RU^{D} represent the downward and upward ramp rates of the diesel generator. The variables $P_t^{\mathrm{B,c}}$, $P_t^{\mathrm{B,d}}$, and E_t^{B} correspond to the battery's charging power, discharging power, and state of charge (SoC). Additionally, $\eta^{\mathrm{B,c}}$ and $\eta^{\mathrm{B,d}}$ indicate the efficiency of the battery during the charging and discharging processes. ε denotes the rate at which the battery discharges on its own. $\underline{E}^{\mathrm{B}}$ and $\bar{E}^{\mathrm{B}}$ represent the minimum and maximum state of charge (SoC) limits of the battery. $\bar{P}^{\mathrm{B}}$ indicates the maximum power capacity of the battery. ξ_t^{L} refers to the power consumed by the load, while P_t^{R} signifies the power supplied by renewable energy sources (RES).

$$\min_x G(\boldsymbol{x}, \boldsymbol{\xi}) = \sum_{t \in \boldsymbol{\Omega}_T} \left(C_t^{\mathrm{L}} + C_t^{\mathrm{D}} + C_t^{\mathrm{B}} + C_t^{\mathrm{H}} + C_t^{\mathrm{aging}} \right) \tag{37}$$

$$C_t^{\mathrm{L}} = c^{\mathrm{L}} P_t^{\mathrm{L}} \Delta t, C_t^{\mathrm{D}} = c^{\mathrm{D}} P_t^{\mathrm{D}} \Delta t, C_t^{\mathrm{B/H}} = c^{\mathrm{B/H}} P_t^{\mathrm{B/H,d}} \Delta t \tag{38}$$

$$\text{Constraints}: \forall t \in \boldsymbol{\Omega}_T \tag{39}$$

$$\underline{p}^{\mathrm{D}} \leq P_t^{\mathrm{D}} \leq \bar{P}^{\mathrm{D}}$$

$$-RD^{\mathrm{D}} \leq P_{t+1}^{\mathrm{D}} - P_t^{\mathrm{D}} \leq RU^{\mathrm{D}} \tag{40}$$

$$0 \leq P_t^{\mathrm{L}} \leq \xi_t^{\mathrm{L}} \tag{41}$$

$$0 \leq P_t^{\mathrm{R}} \leq \xi_t^{\mathrm{R}} \tag{42}$$

$$0 \leq P_t^{\mathrm{G}} \leq \bar{P}_t^{\mathrm{G}} \tag{43}$$

$$P_t^{\mathrm{G}} + P_t^{\mathrm{R}} + \left(P_t^{\mathrm{B,d}} - P_t^{\mathrm{B,c}}\right) + \left(P_t^{\mathrm{H,d}} - P_t^{\mathrm{H,c}}\right) + P_t^{\mathrm{L}} + p_t^{ev} = \xi_t^{\mathrm{L}} \tag{44}$$

The economic dispatch of a microgrid over multiple time periods is outlined in the following section. Subsequently, we will detail the approach used to address this issue through both offline and online optimization stages.

$$\min_x G\left(\boldsymbol{x}, \boldsymbol{\xi}\right) \tag{45}$$

$$\text{s.t.} \quad (12)-(18),\ (19)-(23),(24)-(35),(39)-(44) \tag{46}$$

Coordinated optimization framework without reliance on predictions

We introduce a coordinated optimization approach structured in two phases, as shown in Fig. 2, comprising both pre-planned and real-time optimization steps. The initial offline phase establishes reference points for the state of charge (SoC) in hydrogen storage systems by analyzing uncertainty sets on renewable energy sources. These benchmarks reduce the risk of short-sighted choices, continuously refined using kernel regression as fresh data becomes available. In the subsequent online phase, decisions are dynamically optimized by a novel online convex optimization (OCO) technique, eliminating the need for forecast data.

The first-stage optimization determines robust annual SoC references for hydrogen storage, accounting for the worst-case uncertainty scenarios. These references are not static; they serve as initial baselines that are dynamically adjusted in the second stage based on incoming real-time data. This design ensures a seamless transfer of strategic planning into real-time control.

Offline-stage

In real-world situations, the scheduling of microgrid operations is usually planned a day ahead of execution, resulting in numerous uncertainties that complicate the planning process. Considering the defined sets of uncertainty, the development of the model based on two-stage robust optimization can be articulated as follows:

$$\min_{x_t} \sum_{t\in T} a_t^T x_t + \max_{U_{RE}} \min_{y_t \in O(x_1,\dots,x_T,g)} \sum_{t\in T} b_t^T y_t \tag{47}$$

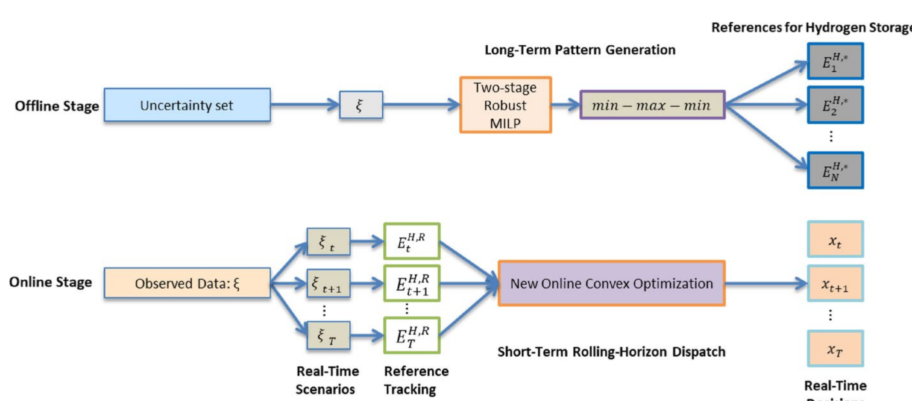

Fig. 2 A schematic representation of a dual-stage optimization framework that operates independently of predictive modeling

$$C_t x_t \leq c_t \tag{48}$$

$$O\left(x_1, \ldots, x_T, g\right) = \left\{ y_t \middle| \begin{array}{c} H_t y_t \leq h_t \\ G_t y_t \leq L_t d_t - i_t x_t - E_t x_t - K_t g_t \\ N_t y_t \leq w - M_t x_t \end{array} \right\} \tag{49}$$

In this model, x and y are the variables that determine the system's operational choices. Coefficients C, E, G, H, K, L, M, and N provide additional support, and parameters a, b, c, h, and w act as supporting values within the simplified robust framework. The variable g represents the uncertain output of renewable energy sources. The two-stage robust model presented here initially focuses on establishing a secure energy plan, covering distributed generation sources, energy allocations for storage systems, electric vehicle energy planning, power management for both battery and hydrogen reserves, as well as adaptable demand scheduling. In this phase, the goal is to reduce the microgrid's overall costs by focusing on scenarios that represent the most challenging outcomes. After determining the initial decisions, a model is applied to simulate the microgrid's response under uncertain conditions, prioritizing resilience against adverse events. The optimization task structured in Eqs. (47)–(49) follows a min- max- min framework that commercial software tools are unable to handle effectively. As a result, the robust model presented is tackled using a C&CG algorithm, which is thoroughly discussed in [21] for its efficacy in such cases.

$$\min_{x_t, \varphi} \varphi \tag{50}$$

$$C_t x_t \leq c_t \tag{51}$$

$$\varphi \geq \sum_{t \in T} a_t^T x_t + \sum_{t \in T} b_t^T y_t^\vartheta \tag{52}$$

$$H_t y_t^\vartheta \leq h_t \tag{53}$$

$$G_t y_t^\vartheta \leq L_t d_t^\vartheta - i_t^\vartheta x_t - E_t x_t - K_t g_t^\vartheta \tag{54}$$

$$N_t y_t^\vartheta \leq w - M_t x_t \tag{55}$$

In this context, φ functions as a support variable, while ϑ denotes the iteration count. The main aim of the master problem is to determine the best possible decision for the first stage, factoring in the most challenging scenarios identified within the sub-problem. As a result, the master problem sets a foundational lower limit for the two-stage robust optimization model outlined in Eqs. (47)–(49). The secondary problem is crafted to determine the most adverse scenario within the defined bounds of uncertainty. In particular, for the robust optimal decisions established in the initial stage and symbolized by x_t^*, this sub-problem is structured as presented:

$$\eta\left(x^*\right) = \max_{g \in U_{RE}} \min_{y_t \in L(x_1, \ldots, x_T, g)} \sum_{t \in T} b_t^T y_t \tag{56}$$

$$H_t y_t \leq h_t \quad (\pi_t) \tag{57}$$

$$G_t y_t \leq L_t d_t - i_t x_t^* - E_t x_t^* - K_t g_t \quad (\gamma_t) \tag{58}$$

$$N_t y_t \leq w - M_t x_t^* \ \ (\omega_t) \tag{59}$$

In this context, η functions as an additional variable, and π, γ, and ω are the dual variables corresponding to constraints (57)–(59) respectively. This *max-min* formulation provides an upper limit for the two-stage model that employs robust optimization, as presented in Eqs. (47)–(49). Nonetheless, optimizing this problem directly is impractical. Fortunately, because the sub-problem maintains linear properties, duality theory can be applied to transform it into the following equivalent form:

$$\max_{\pi,\gamma,\omega,g} \sum_{t\in T} \left\{ h_t^T \pi_t + (L_t d_t - I_t x_t^* - E_t x_t^* - K_t g_t)^T \gamma_t + (w + M_t x_t^*)^T \omega_t \right\} \tag{60}$$

$$H_t^T \pi_t + G_t^T \gamma_t + N_t^T \omega_t = b_t \tag{61}$$

$$\pi_t \leq 0,\ \gamma_t \leq 0,\ \omega_t \leq 0\,, d_t \in U_D^p,\ i_t \in U_D^q, g_t \in U_{RE} \tag{62}$$

Once the equivalent formulation for the subproblem has been addressed, the most unfavorable outcome is assessed across the uncertainty ranges during each iteration, and the relevant constraints (52)–(55) are incorporated into the main problem. The iterative procedure concludes once the gap between the maximum and minimum limits falls below a specified threshold. The flowchart in Fig. 3 visually represents the suggested solution algorithm. This two-tier model is established by separating the optimization challenge into a primary problem and a secondary one, as shown in the flowchart provided in Fig. 3. The primary aim of this model is to manage microgrids, considering the uncertainties involved.

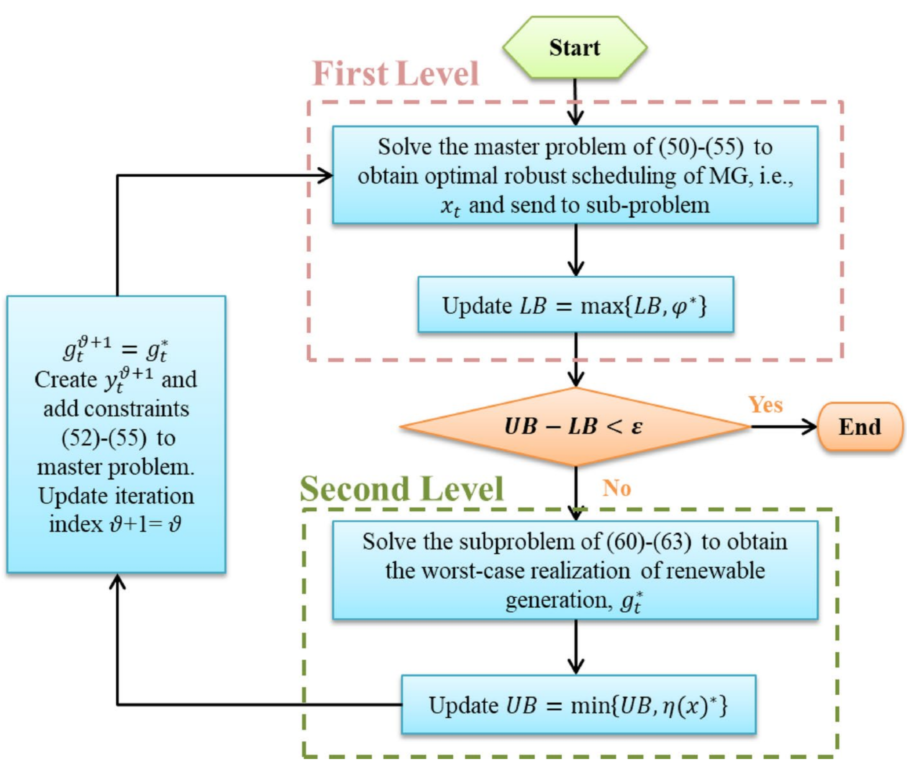

Fig. 3 Diagram illustrating the suggested algorithm for the two-phase solution

To terminate the algorithm, a convergence threshold ϵ is introduced to control the acceptable optimality gap between the upper and lower bounds. In this study, the value of ϵ is set to 0.001 (or 0.1% of the total cost), which provides a practical balance between accuracy and computational time. The algorithm stops when the relative gap satisfies:

$$\frac{UB - LB}{|LB|} \leq \epsilon \tag{63}$$

where UB and LB denote the upper and lower bounds, respectively, obtained during the iteration process. While a smaller value of ϵ can yield more precise solutions, it also increases the number of iterations and overall runtime. Therefore, this threshold should be tuned based on the problem scale and available computational resources.

Online-stage

We introduce a novel approach for reference tracking that utilizes data-driven techniques, integrating the 'lookback' methodology from past datasets and the 'lookahead' methodology from recent observations. To start, we denote $\boldsymbol{\xi}_{[t]}$ as the sequence of uncertainties observed from the initial time slot up to the current time slot t, as indicated in (64). Furthermore, $\boldsymbol{\xi}_{s,[t]}$, outlined in (65), denotes the historical sequence of uncertainties relevant to scenario s. Next, the dynamic weights $\omega_{s,t}$ for each historical scenario are determined by evaluating the resemblance between $\xi_{[t]}$ and $\xi_{s,[t]}$. This process utilizes a Gaussian kernel function in conjunction with Euclidean distance, as described in (66). To determine an appropriate value for the bandwidth parameter σ in the Gaussian kernel function, a data-driven tuning procedure is employed. Specifically, we apply a simple cross-validation strategy using a rolling historical window. A range of candidate σ values is evaluated, and the one that minimizes the prediction error on a validation subset is selected for use in the real-time kernel regression model. This ensures that the kernel width is well-matched to the structure of the input data and avoids the risks of both overfitting (when σ is too small) and oversmoothing (when σ is too large). While this heuristic tuning performs well in our experiments, we note that adaptive bandwidth selection techniques could further improve generalization and robustness in more complex or dynamic settings, which will be considered in future research. Additionally, to reflect the changes over time, the Gaussian kernel function incorporates a scaling factor t. The ideal bandwidth σ can be determined using heuristic techniques, including the bisection approach. Furthermore, the weights are adjusted during real-time operations rather than relying on average or heuristic estimates. The reference for the state of charge (SoC) of hydrogen storage is revised as (67). This revised reference plays a crucial role in defining the efficiency range of hydrogen storage, thereby addressing the nonconvexity challenges associated with convex optimization methods.

$$\boldsymbol{\xi}_{[t]} = \left\{\xi_1^G, \xi_1^L, \xi_1^R, \ldots, \xi_t^G, \xi_t^L, \xi_t^R\right\} \tag{64}$$

$$\xi_{s,[t]} = \left\{\xi_{s,1}^{\mathrm{G}}, \xi_{s,1}^{\mathrm{L}}, \xi_{s,1}^{\mathrm{R}}, \ldots, \xi_{s,t}^{\mathrm{G}}, \xi_{s,t}^{\mathrm{L}}, \xi_{s,t}^{\mathrm{R}}\right\} \tag{65}$$

$$\omega_{s,t} = \frac{K_t\left(\xi_{[t]}, \xi_{s,[t]}\right)}{\sum_{s'=1}^{N} K_t\left(\xi_{[t]}, \xi_{s',[t]}\right)}, K_t(x,y) = e^{-\frac{(\| x-y\|_2)^2}{t\sigma^2}} \tag{66}$$

$$\boldsymbol{E}_{[t]}^{\mathrm{H,R}} = \sum_{s=1}^{N} \omega_{s,t} \boldsymbol{E}_{s,[t]}^{\mathrm{H},*} \tag{67}$$

A formulation for real-time corrective dispatch is presented in (68), designed to reduce immediate operational expenses while ensuring that the state of charge (SoC) of hydrogen storage aligns with the reference value. The penalty coefficient φ regulates the extent of deviation from this reference SoC. This can be succinctly expressed in (69). The functions f_t and g_t denote the dynamic objective function and constraints that fluctuate over time, influenced by the reference state of charge for hydrogen storage, $E_t^{\mathrm{H,R}}$, and associated uncertainties, ξ. Utilizing Lagrangian Relaxation enables us to achieve the optimal solution as indicated in Eq. (70). λ_t represents the dual variables associated with the constraints g_t. The notation $\langle x, y\rangle$ refers to the typical inner product. Nevertheless, in the absence of information regarding uncertainties ξ, the online decision-maker lacks knowledge of f_t and g_t. Therefore, we will develop a new online convex optimization algorithm to address this problem.

$$\begin{aligned} &\min_{x_t} G\left(\boldsymbol{x}_t, \boldsymbol{\xi}_t\right) + \varphi\left(E_t^{\mathrm{H}} - E_t^{\mathrm{H,R}}\right)^2 \\ &\text{s.t.} \quad (12) - (18)\,,\ (19) - (23)\,, (24) - (35)\,, (39) - (44) \end{aligned} \tag{68}$$

$$\min_{x} f_t\left(x_t\right) \text{ s.t. } g_t\left(x_t\right) \leq 0 \tag{69}$$

$$x_t = \underset{x}{\operatorname{argmin}} \left\{ f_t(x) + \langle \lambda_t, g_t(x) \rangle \right\} \tag{70}$$

The refined SoC reference $\boldsymbol{E}_{s,[t]}^{\mathrm{H},*}$ (see Eq. (67)) directly incorporates the outcome of the offline stage. The online convex optimization (OCO) algorithm then minimizes the deviation from this reference (Eq. (69)), effectively embedding the strategic offline decision into each real-time adjustment. This coupling ensures that the real-time decisions remain aligned with long-term objectives while allowing flexibility to adapt to short-term uncertainties.

New OCO algorithm

The primary principle of new-OCO is to leverage historical information to gauge the current conditions. A virtual queue serves as a proxy for the unknown dual variables. Therefore, we establish the update strategy for the virtual queue, decision-making processes, and associated weights as described in Eqs. (71)–(73). In conclusion, the weighted average for the dispatch decision can be derived as presented in (74). The details of the new-OCO algorithm and the complete two-stage coordinated optimization framework are outlined in Algorithm 2.

$$Q_{i,t-1} = Q_{i,t-2} + \beta_{t-1}\left[g_{t-1}\left(x_{t-1}\right)\right]_+ \tag{71}$$

$$x_{i,t} = \arg\min_x \left\{ \alpha_{i,t-1}\langle \partial f_{t-1}\left(x_{t-1}\right), x\rangle + \alpha_{i,t-1}\beta_{i,t-1}\langle Q_{i,t-1}, \left[g_{t-1}(x)\right]_+\rangle + \|x - x_{i,t-1}\|^2 \right\} \tag{72}$$

$$\updownarrow_{i,t-1} = \langle \partial f_{t-1}\left(x_{t-1}\right), x_{i,t-1} - x_{t-1}\rangle, \rho_{i,t} = \frac{\rho_{i,t-1} e^{-\gamma \updownarrow_{i,t-1}}}{\sum_{i=1}^{N} \rho_{i,t-1} e^{-\gamma \updownarrow_{i,t-1}}} \tag{73}$$

$$x_t = \sum \rho_{i,t} x_{i,t}{}_{t=1}^{N} \tag{74}$$

First Stage: Offline Optimization
Input: Uncertainty sets of RES ξ
Output: Reference for hydrogen storage $\boldsymbol{E}_N^{\mathrm{H},*}$
  for $S = 1$ **to** N **do**
    Solve the two-stage robust MILP problems (45)-(46) to generate the SoC references of hydrogen storage.
  end
Second Stage: Online Optimization
Input: Historical reference for hydrogen storage $E_s^{\mathrm{H},*}$; Real-time observation of RES and load ξ_t
Output: Real-Time Dispatch Decisions x_t
Step 1 –Initialization
    Set $Q_{i,t} = 0, x_{i,1} \in X, x_1 = \sum_{i=1}^{N} \rho_{i,1} x_{i,1}$
    $\rho_{i,1} = (M+1)/[i(i+1)M], \forall i \in \{1,2,\cdots,M\}$
Step 2 - Reference Tracking & new-OCO
  for $t = 2$ **to** T **do**
    Update real-time SoC reference as (64)-(67);
    for $i = 1$ **to** M **parallel do**
      Update virtual queue $Q_{i,t}$ as (71);
      Update decisions $x_{i,t}$ as (72);
      Update weights $\rho_{i,t}$ as (73).
    end
    Calculate the dispatch decision x_i as (74).
  end

Algorithm 2 An online optimization method employing a dual-Phase approachwithout relying on forecasting.

Note 1 (Estimation). The function $f_t(x)$ is approximated using the linearization provided by the first-order Taylor series $\langle \partial f_{t-1}(x_{t-1}), x \rangle$. The variable λ_t is replaced with a conceptual queue represented as $Q_{i,t-1}$. The constraint function $g_t(x)$ is modified to a clipped version, noted as $[g_{t-1}(x)]_+$. To ensure that the optimization problem remains convex and to facilitate better convergence of the algorithm, a regularization term in the form of $|x - x_{i,t-1}|^2$ is introduced.

Note 2 (Simultaneous Learning). The selection of an appropriate learning rate (step size) is crucial but often presents difficulties. We implement distinct learning rates for the initial two components, $\alpha_{i,t-1}$ and $\beta_{i,t-1}$. Instead of relying on constant or adaptive rates, we utilize the expert-tracking algorithm outlined in reference [29]. This algorithm calculates x_t simultaneously while applying different learning rates, as illustrated in Eq. (72). The weights associated with each expert, $\rho_{i,t}$, are adjusted according to their observed performance using an exponential approach, as depicted in Eq. (73).

Note 3 (Updates to the Virtual Queue). As outlined in our earlier research [29], the dual variables related to the long-term constraints stay constant unless the optimum meets the constraint limits. When the optimum does meet these limits, the dual variables escalate, indicating an imposed penalty. The adjustment of the virtual queue adheres to the methodology detailed in Eq. (71) to control violations of the constraints.
OCO emphasizes analyzing regret (Reg), defined in (75), where y_t represents the best possible solution globally. A range of OCO methods are developed to ensure that Reg remains a sublinear function of T by optimizing parameters and updating strategies. This suggests that, in retrospect, the algorithm's efficiency approaches that of the global optimum as T increases indefinitely. Following this, we will discuss the specific parameter configurations and provide a proof to confirm the achievement of strictly sublinear dynamic regret.

$$\mathrm{Reg} = \sum_{t=1}^{T} \left[f_t\left(x_t\right) - f_t\left(y_t\right) \right] \tag{75}$$

Assumption 1 Both functions f_t and g_t exhibit convex characteristics. The feasible set X is not only convex but also closed, with a finite diameter denoted as $d\left(\boldsymbol{X}\right)$.

$$\| x - y \| \leq d\left(\boldsymbol{X}\right), \forall x, y \in \boldsymbol{X} \tag{76}$$

Assumption 2 A positive constant, denoted by F, is assumed to exist such that it satisfies this condition.

$$\left| f_t\left(x\right) - f_t\left(y\right) \right| \leq F, || g_t\left(x\right) || \leq F, \forall t \in \boldsymbol{\Omega}_{\boldsymbol{T}}, \forall x, y \in X \tag{77}$$

Assumption 3 The subgradients $\partial f_t\left(x\right)$ and $\partial g_t\left(x\right)$ are assumed to be present. Additionally, a positive constant G can be identified, ensuring their boundedness.

$$|| \partial f_t\left(x\right) || \leq G, || \partial g_t\left(x\right) || \leq G, \forall t \in \boldsymbol{\Omega}_{\boldsymbol{T}}, \forall x, y \in \boldsymbol{X} \tag{78}$$

Theorem 1 *Under the conditions specified in Assumptions1–3 and with parameters defined in Eq. (79), where κ falls within the range [0,c], c is a value in the interval (0,1), and* α_0, β_0, *and* γ_0*are positive constants with* γ_0*in the range (0,1/(√2G)), we can express the performance metrics for Reg and Vio as shown in Eq. (80).*

$$M = \lfloor \kappa \log_2(1+T) \rfloor + 1, \alpha_{i,t} = \frac{\alpha_0 2^{i-1}}{t^c}, \beta_{i,t} = \frac{\beta_0}{\sqrt{\alpha_{i,t}}}, \gamma = \frac{\gamma_0}{T^c} \tag{79}$$

$$\mathrm{Reg} = \mathcal{O}\left(T^c(1+P_x)^{1-\kappa} + T^{1-c}(1+P_x)^{\kappa}\right) \tag{80}$$

Simulation results

Table 3 outlines the essential parameters and system configurations. The battery and hydrogen storage capacities are each designed to be 50% of the overall load capacity. The energy storage times for the battery and hydrogen systems are 2 h and 400 h, respectively. The total capacity of the renewable energy sources installed is 200 kW, split evenly between solar and wind technologies. Information about the cost coefficients is available in reference [30]. We illustrate the efficacy of our proposed approach through analysis of two distinct datasets. First, for the baseline case study, we utilize 15-minute interval historical data concerning solar energy, wind energy, and load spanning from 2014 to 2023, sourced from Elia, the transmission system operator in Belgium. Second, we analyze hourly historical records of wind energy and load covering the period from 1981 to 2020 in North China, which helps to highlight the effects of data resolution and volume.

Table 3 Parameters and configuration of the test microgrid

Parameters	Value	Parameters	Value	Parameters	Value
Initial SoC	0.5	$\bar{P}^{\mathrm{H}}$	50 kW	c^{L}	$\$5/\mathrm{kWh}$
$\eta^{\mathrm{B,c/d}}$	0.9	$\bar{E}^{\mathrm{H}}$	20 MWh	$\bar{P}^{\mathrm{D}}$	50 kW
ϵ	1%/ month	c^{B}	$\$0.02/\mathrm{kWh}$	Wind Capacity	100 kW
$\bar{P}^{\mathrm{B}}$	50 kW	c^{H}	$\$0.03/\mathrm{kWh}$	Solar Capacity	100 kW
$\bar{E}^{\mathrm{B}}$	100 kW	c^{D}	$\$0.3/\mathrm{kWh}$	Load Capacity	100 kW

In this study, we assume $c_{\text{deg}} = 100$ USD/MWh based on a battery replacement cost of 400 USD/kWh and 4,000 cycle lifetime. This value can be adjusted for different battery chemistries. The optimization process is implemented using Julia in conjunction with the JuMP interface, and it is resolved utilizing the Gurobi 11.0 solver. The computations are carried out on a system powered by an Intel Core i7 processor running at 2.20 GHz, equipped with 16 GB of RAM. To initialize the simulation environment, the load at time step $t = 1$ is set based on the first observed value from the historical dataset used for each scenario. This approach ensures that the model begins its operation from a realistic system state. For the Elia (Belgium) dataset, the initial load is 684 kW, while for the North China region dataset, the initial load is 802 kW. These values are directly taken from the real data used in the respective test cases and are held constant across all comparative simulations to maintain consistency.

Offline optimization results

In this section, the study examines two distinct operational modes: offline optimization and online optimization. Each mode is implemented to explore its respective performance in managing energy flow within the microgrid, particularly regarding cost-effectiveness, efficiency, and adaptability in dynamic scenarios. Offline optimization serves as a preliminary step, where annual state-of-charge targets and strategies for the hybrid hydrogen-battery storage system are determined based on uncertainty sets and predefined goals. This offline process provides a structured baseline, leveraging semi-empirical modeling to enhance accuracy in initial planning without requiring real-time data adjustments. On the other hand, the online optimization mode dynamically refines the offline strategies in real time, adapting to fluctuations in load demand, renewable energy supply, and operational conditions. By utilizing kernel regression techniques, the online optimization mode can fine-tune the energy management process, ensuring higher responsiveness and robustness against unpredictable factors. A new online convex optimization method further supports this mode by enabling immediate decision-making adjustments, thus minimizing operational costs and enhancing system reliability. The results obtained from each mode are analyzed and compared to highlight the strengths and limitations of offline versus online optimization in microgrid energy management. The comparative analysis aims to demonstrate the superiority of the combined approach, where offline optimization establishes a solid foundation while online optimization ensures adaptability and responsiveness, ultimately achieving a more efficient and resilient energy system.

Tables 3 and 4 present the results of the offline optimization stage applied to two distinct datasets: the Elia dataset and data from North China, both for the year 2023. These tables provide a comparative view of the microgrid's cost performance, diesel generator power usage, and loss of load power in each region, highlighting the operational efficiency and economic impact of offline optimization. Table 4 shows the offline optimization results for the Elia dataset. The microgrid achieved a total operational cost of $91,250, with the diesel generator producing 309 MWh of power. Importantly, there was no loss of load power, indicating a high reliability level in meeting energy demand

Table 4 Offline two-stage robust optimization of the microgrid in 2023 (Elia dataset)

Cost ($)	Diesel generator power (MWh)	loss of load power (MWh)
91,250	309	0.00

Table 5 Offline two-stage robust optimization of the microgrid in North China

Cost ($)	Diesel generator power (MWh)	loss of load power (MWh)
872,541	413	1.27

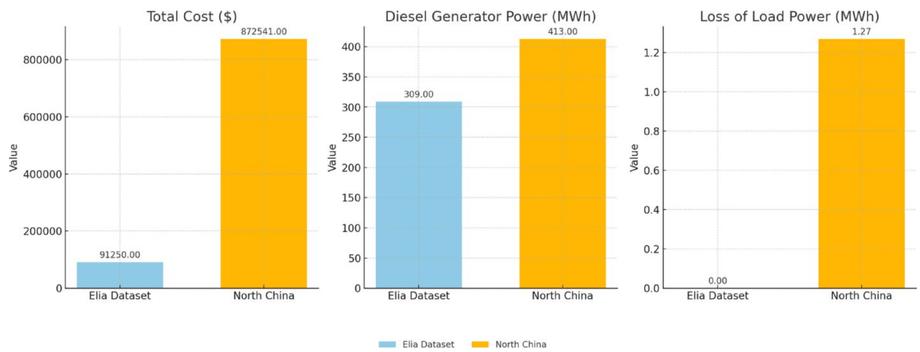

Fig. 4 Offline optimization results

without supply interruptions. Table 5 outlines the offline optimization results for North China. Here, the total operational cost reached $872,541, with a diesel generator output of 413 MWh. The system experienced a minor loss of load power of 1.27 MWh, indicating a slight gap in meeting demand under offline optimization conditions.

These results illustrate the regional differences in cost and reliability when implementing offline optimization. The Elia dataset shows better cost-efficiency and reliability, with no unmet load, while the North China dataset demonstrates slightly higher costs and a minor load deficit. This analysis underscores the need for further refinement through online optimization to achieve enhanced adaptability and efficiency across varied operating conditions.

Figure 4 illustrates the comparative results of the offline optimization stage for two datasets: Elia and North China. This figure breaks down three key performance indicators—Total Cost, Diesel Generator Power, and Loss of Load Power—across separate panels, providing a clear view of regional differences in energy management outcomes. In terms of Total Cost, North China shows substantially higher operational expenses compared to the Elia dataset, indicating potential cost challenges specific to the region. Diesel Generator Power usage is also more prominent in North China, reflecting a greater dependency on conventional power sources, which may be attributed to variations in renewable availability or demand peaks. Finally, Loss of Load Power is minimal in both cases, though North China experiences a slight unmet demand, suggesting room for improvement in real-time adjustments. This figure emphasizes the unique operational characteristics and optimization needs for each region within the microgrid framework.

Figure 5 illustrates the power distribution during the offline optimization phase of the microgrid. This diagram highlights the contributions of various energy sources and storage systems, including the hydrogen storage system, battery, diesel generator, and net load. In this figure, the power output from the hydrogen storage system is depicted, showcasing its role in providing renewable energy and enhancing grid stability. The battery's contribution is also illustrated, representing its function in energy storage and load leveling, which helps manage fluctuations in energy demand. The diesel generator's power output is displayed as well, reflecting its role as a conventional backup source, particularly during peak demand periods or when renewable generation is insufficient.

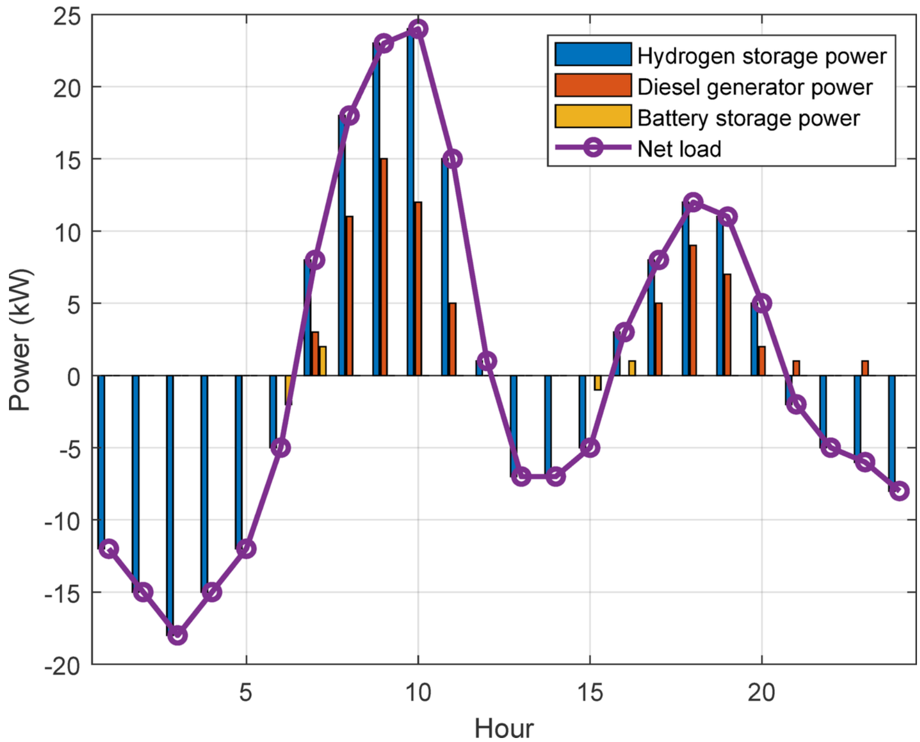

Fig. 5 Power dispatch in offline optimization strategy (Elia dataset)

Table 6 Online convex optimization of the microgrid in 2023 (Elia dataset)

Cost ($)	Diesel generator power (MWh)	loss of load power (MWh)	Time (ms)
121,870	317	4.06	41.02

Finally, the net load is represented, indicating the total energy demand that the microgrid needs to satisfy. This comprehensive view of power distribution underlines the intricate interplay between various generation and storage technologies, emphasizing the importance of optimizing these resources for effective energy management within the microgrid framework.

Online optimization results

The online optimization results provide insight into the real-time operation and decision-making within the microgrid environment. This section compares the Elia and North China datasets to assess performance across key metrics such as total cost, diesel generator power, loss of load power, and computation time. Online optimization allows the microgrid to adapt dynamically to fluctuations in demand and renewable generation, although this flexibility comes with its own set of challenges. By analyzing these metrics, the study aims to highlight areas where each region can improve in terms of efficiency, reliability, and operational costs.

Table 6 summarizes the results of online convex optimization for the Elia dataset in 2023. The total operational cost is $121,870, reflecting the expense incurred in managing real-time adjustments. Diesel generator power usage is 317 MWh, indicating a moderate reliance on conventional generation to support grid stability. Additionally, there is a small loss of load power of 4.06 MWh, meaning that nearly all demand is met with

Table 7 Online convex optimization of the microgrid in North China

Cost ($)	Diesel generator power (MWh)	loss of load power (MWh)	Time (ms)
1,035,471	417	197	48.11

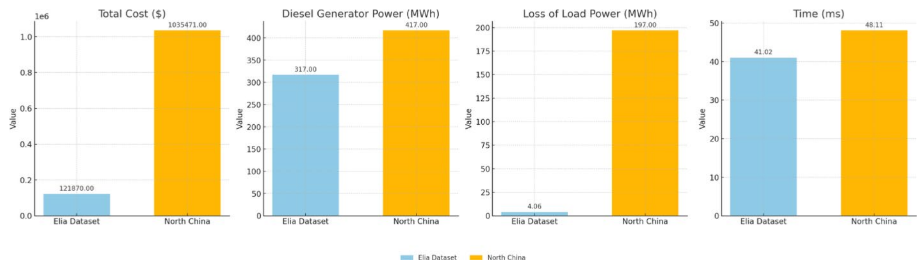

Fig. 6 Online optimization results

minimal interruption. The computation time required for real-time optimization is 41.02 milliseconds, demonstrating that the Elia dataset achieves fast computational performance, essential for effective online management.

Table 7 presents the online convex optimization results for North China. The total cost is significantly higher at $1,035,471, suggesting that operating in North China incurs substantially greater expenses. Diesel generator power usage is also higher at 417 MWh, indicating a heavier reliance on conventional power. Notably, the loss of load power is substantial, reaching 197 MWh, which indicates that North China faces significant challenges in meeting demand through online optimization alone. The computation time is slightly longer at 48.11 milliseconds but remains efficient for real-time processing needs.

These tables highlight the main differences in online optimization performance across regions. North China shows higher operational costs, increased diesel generator use, and a significant unmet demand, indicating that the microgrid there faces greater challenges in maintaining efficient and reliable service.

Figure 6 offers a comparative analysis of the online convex optimization results for the Elia and North China datasets, presenting four distinct metrics: total cost, diesel generator power, loss of load power, and computation time. Each metric is displayed in a separate panel, allowing for a straightforward side-by-side comparison between the two datasets, which reveals significant differences in their operational profiles. In the first panel, we see the total cost of operation for each dataset. North China's operational cost stands out, reaching $1,035,471, which is substantially higher than Elia's $121,870. This discrepancy suggests that North China's operating environment demands more resources and faces higher energy or fuel expenses, possibly due to an increased need for conventional generation. This result points to a potential financial challenge in real-time optimization, as high costs could limit operational efficiency. The second panel illustrates diesel generator power usage. Here, North China shows greater dependence on conventional energy sources, consuming 417 MWh, compared to Elia's 317 MWh. This heavier reliance on diesel power suggests that North China might experience less availability of renewable energy or more demand peaks that conventional generation must support. This dependence may highlight an area for improvement, such as increased renewable integration or enhanced storage options. The third panel, showing loss of load power, reveals one of the most significant differences between the datasets. North China experiences an unmet demand level of 197 MWh, while Elia's is only 4.06 MWh. This high loss of load in North China indicates that the system struggles to meet demand consistently with the current online optimization approach. This unmet demand highlights a critical need for more flexible or responsive optimization strategies that could address this issue and enhance reliability. The final panel, focusing on computation time, shows that both datasets achieve efficient processing times for real-time applications. Elia's computation time is 41.02 milliseconds, while North China's is slightly higher at 48.11 milliseconds. Although this increase is relatively minor, it points to a slightly greater computational requirement for North China, which may stem from the more complex demand and resource dynamics in that region. Main findings from the analysis reveal several key points. North China incurs significantly higher costs than Elia, reflecting the challenges of managing real-time demands with potentially higher fuel or operational expenses. Diesel generator usage is also higher in North China, suggesting a heavier reliance on conventional power sources, potentially due to either limited renewable availability or greater demand fluctuations. Additionally, North China faces a substantial unmet demand, shown by its high loss of load power, indicating that the current online optimization may require enhancements to improve reliability. Although both datasets achieve fast computation times, North China's slightly higher time suggests an increased computational load due to the complex demand environment. These findings emphasize the necessity for further optimizations in North China, focusing on reducing costs, integrating renewable energy more effectively, and improving real-time responsiveness to ensure demand is met consistently.

Figure 7 illustrates the power distribution during the online optimization mode of the microgrid. This diagram provides a visual representation of the contributions from various energy sources, including the hydrogen storage system, battery, diesel generator, and net load. The figure clearly delineates how power is allocated among these components

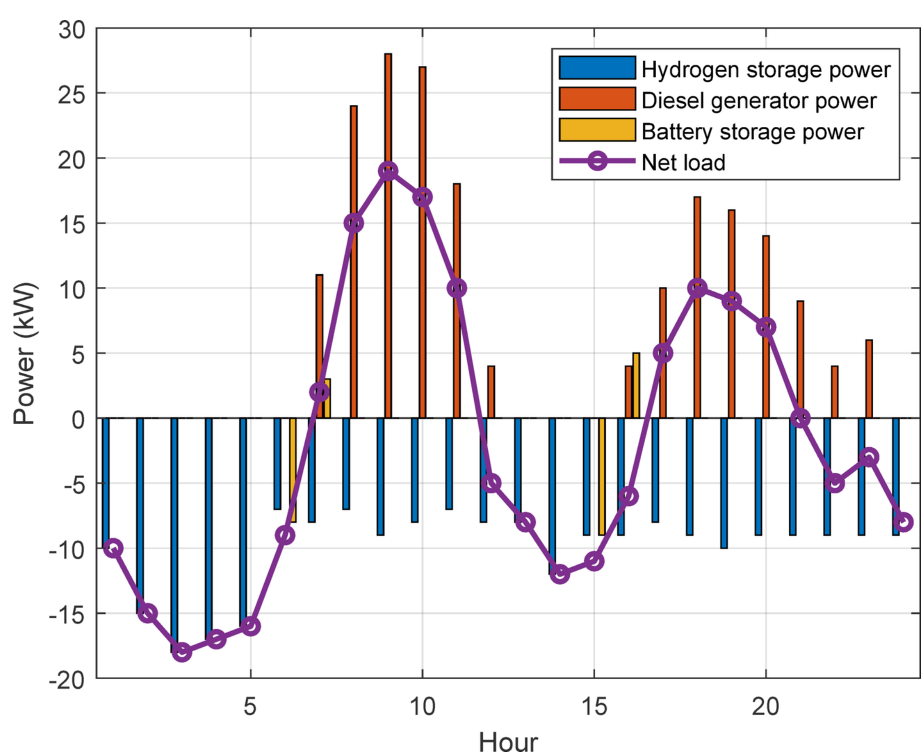

Fig. 7 Power dispatch in online optimization strategy (Elia dataset)

in real-time, reflecting the optimization strategies employed to meet the dynamic energy demands of the microgrid. By showcasing the power output from the hydrogen storage system and battery alongside the diesel generator's contribution, the diagram highlights the integration of renewable energy sources and storage technologies within the microgrid framework. Moreover, the net load curve in the figure indicates the total power demand that must be satisfied by these generation sources. This visualization not only demonstrates the effectiveness of the online convex optimization in balancing power distribution but also underscores the importance of effective energy management in achieving sustainability and reliability within the microgrid system.

Comparing results

In this comparison section, we analyze the outcomes of both offline and online optimization methods applied to a microgrid system using the Elia and North China datasets. By comparing key metrics, we aim to assess the effectiveness and trade-offs of each optimization mode, especially in terms of cost efficiency, diesel generator usage, and load management. Offline optimization generally entails a more pre-planned approach, while online optimization involves real-time adjustments that respond dynamically to changing demands and renewable energy variability. The results indicate substantial differences in performance across these two methods, particularly in terms of operational cost and the ability to meet load requirements.

Figure 8 provides a visual representation of the comparative analysis between offline and online optimization outcomes across three primary metrics: total cost, diesel generator power usage, and loss of load power. Each of these metrics is represented as a separate chart, with bars for both the Elia and North China datasets, allowing for easy side-by-side comparison of offline and online optimization results.

The first panel of Figure 8 presents the total cost associated with each optimization method for both datasets. It shows that, for both Elia and North China, the costs associated with online optimization are significantly higher than those of offline optimization. In the Elia dataset, offline optimization incurs a cost of $91,250, whereas the online method raises this to $121,870. Similarly, North China's costs are considerably higher in both modes, with online optimization reaching $1,035,471 compared to $872,541 for offline. This disparity highlights that online optimization, despite its flexibility, may come with higher operational costs, likely due to the increased reliance on diesel generation and adjustments needed to maintain stability in real time.

The second panel illustrates diesel generator power consumption under each optimization strategy. For the Elia dataset, diesel usage is relatively stable, with only a slight increase in online mode (317 MWh) compared to offline (309 MWh). This trend is

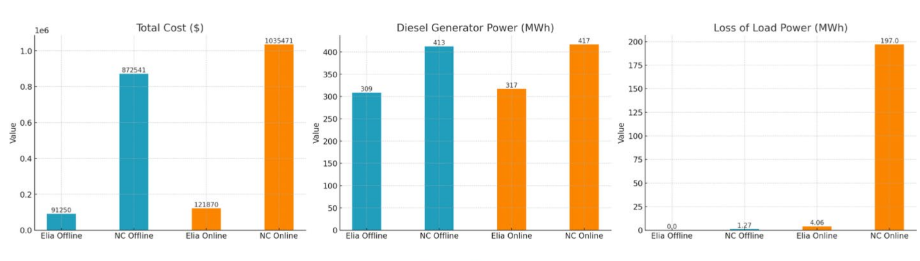

Fig. 8 Comparison of offline and online optimization results

mirrored in the North China dataset, where diesel usage reaches 417 MWh in online mode and 413 MWh in offline. The slight uptick in diesel reliance during online optimization suggests that real-time adjustments require more conventional generation to cover variability in renewable sources or fluctuations in demand, underscoring the need for better integration of renewables to mitigate this dependency.

The third panel displays the loss of load power, which reflects the extent of unmet demand under each optimization mode. The Elia dataset shows a minimal difference between the two modes, with a minor increase from 0 MWh in offline mode to 4.06 MWh in online mode. However, the North China dataset reveals a stark contrast, with offline optimization maintaining a low loss of load power at 1.27 MWh, while online optimization experiences a sharp increase to 197 MWh. This significant unmet demand in North China's online mode indicates that the current real-time optimization approach struggles to fully meet demand, particularly in a high-demand environment. The results imply that while online optimization provides flexibility, its reliability may be compromised, especially in more complex and variable energy scenarios like North China.

The comparative analysis reveals several important findings. Firstly, offline optimization is generally more cost-effective for both regions, suggesting that a planned approach may provide more economical energy management, particularly where renewable availability is variable. Secondly, diesel generator usage tends to be higher in online mode, underscoring a need for better renewable integration and load balancing strategies. Finally, offline optimization appears to manage load more effectively, particularly in the North China dataset, where online optimization results in substantial unmet demand. These insights point to potential improvements for online optimization, focusing on enhancing cost-effectiveness, reducing diesel reliance, and ensuring load consistency to improve overall reliability.

Table 8 provides a comprehensive summary of the key metrics for offline and online optimization results across both the Elia and North China datasets, with a focus on cost, diesel generator power usage, and loss of load power. By comparing the metrics in percentage terms, we can clearly see the differences in performance between the two optimization methods for each dataset. In terms of total cost, offline optimization in the Elia dataset reduces expenses by approximately 25.1% compared to online optimization. This substantial cost savings highlights the efficiency of offline planning in minimizing expenses while maintaining power reliability. For the North China dataset, offline optimization results in a 15.7% reduction in total cost relative to online optimization, showing a considerable cost benefit of offline optimization in a more complex energy environment. For diesel generator usage in the Elia dataset, offline optimization achieves a minor reduction of around 2.5% compared to online optimization. This modest difference suggests that while offline optimization is more cost-effective, it does not greatly reduce the need for diesel generation in this region. In the North China dataset, diesel power usage in offline mode is reduced by about 1.0% relative to online mode. This small

Table 8 Summary comparison of offline and online optimization results for Elia and North China datasets

Metric	Elia Offline	Elia Online	North China Offline	North China Online
Total Cost ($)	91,250	121,870	872,541	1,035,471
Diesel Generator Power (MWh)	309	317	413	417
Loss of Load Power (MWh)	0	4.06	1.27	197

Table 9 Performance comparison between proposed model and predictive scheduling

Metric	Proposed model	Predictive scheduling
Total cost (USD)	7,450	9,800
Load deficit (%)	1.2%	12.5%
Hydrogen utilization (kWh)	3,200	2,400
Battery degradation cost (USD)	160	230

improvement highlights the challenge of reducing conventional power reliance, even with offline optimization, particularly in high-demand settings. In terms of load management, the Elia dataset sees a significant improvement in load reliability with offline optimization, reducing the loss of load power by 100% as it completely eliminates unmet demand, while online optimization results in 4.06 MWh of unmet load. In the North China dataset, offline optimization decreases the loss of load power by 99.4% compared to online optimization. This considerable improvement reflects the offline method's ability to more effectively manage load requirements, even in a region with high variability and demand. Overall, Table 8 highlights the performance advantages of offline optimization in terms of cost, diesel generator usage, and load reliability, with notable improvements in cost reduction and unmet load across both datasets. The data suggests that, while online optimization provides real-time flexibility, offline optimization offers more predictable and cost-effective energy management, particularly for meeting load demand in complex energy environments.

Predictive scheduling benchmark

To provide a fair and comprehensive evaluation of the proposed prediction-independent two-stage robust optimization framework, a predictive scheduling model is implemented as a comparison baseline. The predictive model assumes perfect day-ahead forecasts of renewable generation and load profiles and optimizes the energy dispatch accordingly using a deterministic formulation. This approach reflects the standard method commonly used in practice when reliable forecasts are available, and thus serves as an appropriate benchmark for quantifying the benefits of our robust, forecast-free strategy.

The simulation experiments were conducted using the Elia (Belgium) dataset, which contains hourly renewable energy and load data over a 24-hour horizon. Both models—our proposed robust framework and the predictive scheduling baseline—were tested under identical system configurations, including the same generation capacities, battery and hydrogen storage sizes, EV availability patterns, and cost parameters.

Table 9 presents the detailed performance comparison between the two approaches across four key operational metrics: total operating cost, percentage of unmet load (load deficit), hydrogen energy utilization, and battery degradation cost.

As shown in Table 3, the proposed robust framework outperforms the predictive scheduling model across all performance dimensions.

The proposed model achieves a total cost of 7,450 USD, which is approximately 24% lower than the 9,800 USD incurred by the predictive model. This cost reduction demonstrates the robustness and adaptability of the optimization framework in handling real-time uncertainty without relying on forecasted values.

One of the most critical indicators of microgrid reliability is the percentage of load that remains unmet. The robust model maintains a very low load deficit of only 1.2%, compared to 12.5% under the predictive method. This substantial improvement indicates the enhanced reliability of the proposed strategy under uncertain conditions.

The robust model achieves 33% higher hydrogen energy usage (3,200 kWh vs. 2,400 kWh). This is attributed to the two-stage structure, which allows the system to set more efficient SoC reference targets offline and adjust in real time based on actual operating conditions. In contrast, the predictive model underutilizes hydrogen due to its reliance on static schedules based on forecasts.

The proposed model results in a lower battery degradation cost (160 USD), compared to 230 USD in the predictive model. This reduction is due to the more flexible and efficient dispatch of storage resources, which helps to avoid excessive cycling.

Overall, the results confirm that the proposed method provides better cost-efficiency, system reliability, and energy resource management under uncertainty. These benefits demonstrate the importance of robust, prediction-independent optimization strategies, particularly in modern energy systems with high penetration of renewables and variable demand.

Conclusion

This paper explores a comprehensive approach to optimizing the energy flow in a microgrid environment that integrates renewable sources, adaptable consumption, electric vehicles, and a hybrid hydrogen-battery storage system. By implementing a novel two-stage optimization framework, this research establishes both offline and online optimization strategies that manage long-term energy flow while addressing short-term uncertainties in load demand and renewable generation. The proposed approach includes a prediction-independent strategy, where annual state-of-charge targets are determined offline and refined in real time through kernel regression techniques, as well as operational decisions made with an online convex optimization method. A robust two-stage optimization strategy further addresses uncertainties, using a column-and-constraint generation algorithm to enhance reliability. The main findings show that offline optimization is generally more cost-effective and efficient in managing load compared to online optimization. Specifically, the offline approach achieves substantial cost savings and minimizes unmet demand in both the Elia and North China datasets, demonstrating its suitability for maintaining energy stability over long periods. The results indicate that offline optimization can reduce total operational costs by up to 25% in certain cases while ensuring consistent load reliability, especially in regions with high demand variability. Additionally, although online optimization offers flexibility in adjusting to real-time fluctuations, it exhibits higher operational costs and is more reliant on diesel generators to meet immediate demands. The disparity in load management, particularly in the North China dataset, further suggests that offline optimization provides a more predictable solution for high-demand, variable energy environments.

This work provides a foundation for further research in energy optimization within microgrids, with several opportunities for improvement and expansion. One potential direction is to develop hybrid strategies that leverage the cost-effectiveness of offline optimization while incorporating adaptive real-time adjustments to improve load management. While the current framework adopts aggregated models for electric vehicles and a simplified hydrogen system, more detailed modeling of individual EV behavior—such as user availability patterns, mobility constraints, and decentralized charging logistics—as well as key hydrogen processes including compression, storage thermodynamics, and auxiliary energy consumption, can further improve the physical accuracy of the model. However, incorporating such levels of detail would significantly increase the dimensionality and complexity of the optimization problem, which may affect the convergence of the proposed two-stage optimization framework. As such, these enhancements are left for future research. Additionally, future studies could explore the integration of advanced machine learning techniques to further enhance the prediction accuracy of load demand and renewable generation, potentially improving the efficiency and reliability of online optimization for complex energy systems.

List of Symbols

Indices and sets

Ω_I, Ω_P	Configurations for simultaneous learning processes and the segment related to the efficiency of hydrogen storage
Ω_S, Ω_T	Collections pertaining to past situations and corresponding durations
x, ξ	Collections of variables for decision-making and probabilistic factors, respectively
i, p	Metrics pertaining to the simultaneous learning order and the efficiency of hydrogen storage section, respectively
s, t	Metrics pertaining to past scenarios and specific timeframes, respectively

Parameters

$\alpha_{i,t}, \beta_{i,t}, \gamma$	The incremental adjustments utilized in the OCO algorithm, correspondingly
$\eta^{\mathrm{B,c}}, \eta^{\mathrm{B,d}}$	The efficiency of battery storage during the processes of charging and discharging, respectively
$\eta^{\mathrm{H,c}}, \eta^{\mathrm{H,d}}$	The efficiency of hydrogen storage during the charging and discharging processes
$\omega_{s,t}, \varphi$	Weights that adjust based on past scenarios and a penalty factor for maintaining reference tracking
$\bar{E}^{\mathrm{B}}, \bar{E}^{\mathrm{H}}$	The maximum state-of-charge limits for both battery and hydrogen storage systems
$\bar{P}^{\mathrm{B}}, \bar{P}^{\mathrm{H}}$	The maximum power limits for battery and hydrogen storage systems, respectively
$\underline{E}^{\mathrm{B}}, \underline{E}^{\mathrm{H}}$	The minimum state of charge limits for both battery and hydrogen storage systems
$\underline{p}^{\mathrm{B}}$	Minimum power limit of the battery
$\underline{p}^{\mathrm{D}}, \overline{P}^{\mathrm{D}}$	The power limits for the diesel generator are defined by its minimum and maximum output levels
ε	The rate at which battery storage loses charge when not in use
A_p, B_p	The slope and y-intercept of the individual linear segments used for charging are defined
$c^{\mathrm{B}}, c^{\mathrm{H}}, c^{\mathrm{D}}, c^{\mathrm{L}}$	The costs associated with marginal discharges for battery and hydrogen storage, along with the fuel expenses for diesel generators and the pricing for load reduction, are outlined
C_p, D_p	The slope and intercept of the segmented linear discharge components, respectively
$Q_{i,t}$	A digital waiting line designed for the OCO optimization technique
$RD^{\mathrm{D}}, RU^{\mathrm{D}}$	The rates at which the diesel generator can increase or decrease its output power are referred to as the upward and downward ramping rates, respectively

Variables

$E_t^{\mathrm{B}}, E_t^{\mathrm{H}}$	The levels of charge in the battery and the hydrogen storage system, respectively
$h_t^{\mathrm{c}}, h_t^{\mathrm{d}}$	The generation and utilization of hydrogen
$P_t^{\mathrm{B,c}}, P_t^{\mathrm{B,d}}$	The energy output and input of battery storage systems
$P_t^{\mathrm{D}}, P_t^{\mathrm{L}}, P_t^{\mathrm{R}}$	Power generated by diesel generators, the energy lost due to insufficient load capacity, and the renewable energy output that has been allocated, respectively
$P_t^{\mathrm{H,c}}, P_t^{\mathrm{H,d}}$	The power associated with both the charging and discharging processes of hydrogen storage.
$z_t^{\mathrm{c}}, z_t^{\mathrm{d}}$	Variables in binary form representing the charging and discharging phases of hydrogen storage systems

Author contributions

Mohamad Javad Mohamadi: Conceptualization, Methodology. Mohammad Tolou Askari: Validation, Supervision, Review & Editing of the manuscript, Formal analysis. Mahmoud Samiei Moghaddam: Data curation, Resources, Writing - Original Draft, Software. Vahid Ghods: Conceptualization, Methodology, Visualization.

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Data availability

Availability of data and materialsData is available on request. To request data, contact the corresponding author Mohammad Tolou Askari by e-mail: tolouaskarimohamad@gmail.com at Islamic Azad University, Semnan branch, Semnan, Iran.

Declarations

Competing interests

The authors declare no competing interests.

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